NAME _____ NetID _____

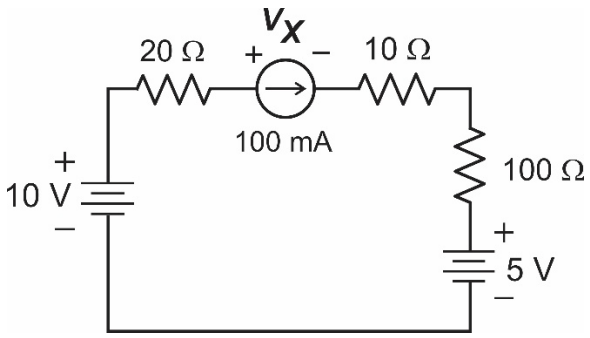
ECE-UY 2004 **SAMPLE MIDTERM EXAM #1** – 3 Problems, 1 Extra Credit **Fall 2026**

READ THESE INSTRUCTIONS

- **RETURN this sheet with your blue book**, **FIVE** points deducted is you fail to return exam sheet.
- Enter your name and NetID. (not N#)
- This is a closed book exam, no notes or paper of any kind – **Reference Sheet Attached.**
- Scrap paper will be provided upon request.
- **SHOW ALL YOUR WORK, NO POINTS FOR JUST AN ANSWER ON ANY STEP**

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1) KVL (10 pts): Use Ohm's Law and KVL to solve, do not use Nodal or Mesh Analysis



A) Which KVL **formula** will you be using in your analysis, are you adding voltage "drops" or voltage "rises" as you progress around the loop?

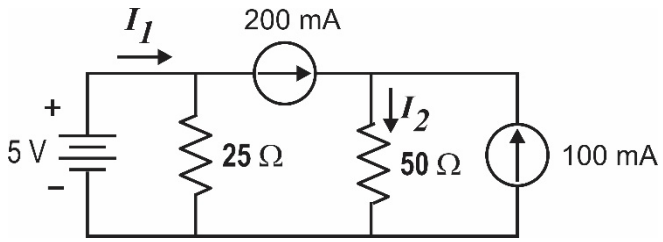
B) Using KVL, find the unknown voltage, V_X (4 pts)

C) Find the power in the three sources, state whether the power is delivered or absorbed. Fill in the table below. (2 pts each)

Component	Power (watts)	Delivered/Absorbed?
10 V		
100 mA		
5 V		

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2) KCL (10 pts): Solve using Ohm's law, KCL, series/parallel combinations and/or power relations. Do not use a formal nodal or mesh analysis procedure.



A) What is the KCL **formula** that you will be using for your analysis (choose one: $\sum I_{IN} = 0$, $\sum I_{OUT} = 0$ or $\sum I_{IN} = \sum I_{OUT}$).

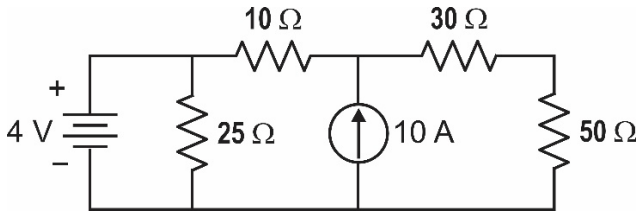
B) Find unknown currents I_1 and I_2. (6 pts)

C) Find the **power** in the ***voltage*** source. Determine whether the source is delivering or absorbing power and briefly explain why (2 pts).

D) Find the **power** in the 100 mA ***current*** source. Determine whether the source is delivering or absorbing power and briefly explain why (2 pts).

Next page for problems 3-4

3) Basics (10 pts)



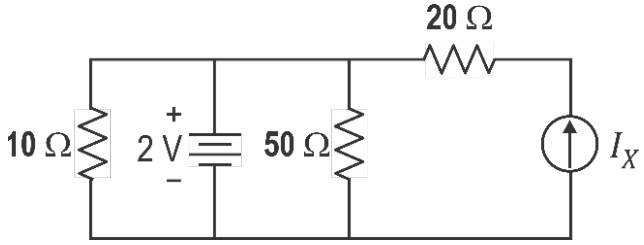
A) List the components that are in **parallel** (3 pts)

B) List the components that are in **series** (3 pts)

C) How many unique **nodes** are in this circuit? (2 pts)

D) How many **loops** are in this circuit? (2 pts)

4) EXTRA CREDIT: KVL/KCL (+2 pts)



What is I_X that creates a condition that the voltage source has no power (P_{2V} = 0 W)? Show your calculation, no points for just an answer.

END OF MIDTERM EXAM 1

DEFINITIONS

J = joule
kWh = kilowatt-hour
hp = horsepower
C = coulomb
s = second
A = ampere ("amp")
V = voltage ("volt")
W = watt

CONSTANTS

Electron Charge = -1.602×10^{-19} C
Proton Charge = $+1.602\times10^{-19}$ C

BASIC CONVERSIONS

1 kWh = 3.6 x 10^6 J
1 hp = 745.7 W
1 A = 1 C/s
1 V = 1 J/C
1 W = 1 J/s = 1 V-A

BASIC UNITS

10^{-12}	pico	p	10^{0}	-	-
10^{-9}	nano	n	10^{3}	kilo	k
10^{-6}	micro	μ	10^{6}	mega	M
10^{-3}	milli	m	10^{9}	giga	G

OHM'S LAW: $V = IR,\ I = {}^{V}/_{R},\ R = {}^{V}/_{I}$

BASIC POWER (DC)

Sources: $P = VI$, sources can deliver or absorb power

Resistors: $P = VI = I^2R = {}^{V^2}/_{R}$, only absorb power

Power Efficiency: $\eta = \frac{P_{OUT}}{P_{IN}} x100\%$

COMPONENTS IN SERIES

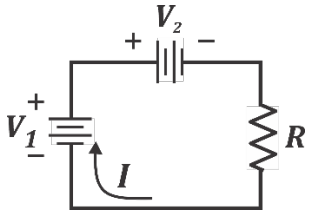
Have the Same Current

COMPONENTS IN PARALLEL

Have the Same Voltage

KIRCHHOFF'S VOLTAGE LAW

$$\sum_{drops}^{loop} \boldsymbol{V_i} = \mathbf{0}$$



EXAMPLE:

$$-V_1 + V_2 + IR = 0$$

OR

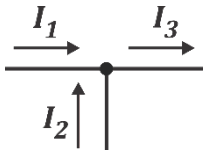
$$\sum_{rises}^{loop} \boldsymbol{V_i} = \mathbf{0}$$

EXAMPLE:

$$+V_1 - V_2 - IR = 0$$

KIRCHHOFF'S CURRENT LAW

$$\sum_{in}^{node} \boldsymbol{I_i} = \mathbf{0}$$



EXAMPLE:

$$I_1 + I_2 - I_3 = 0$$

OR

$$\sum_{in}^{node} \boldsymbol{I_i} = \sum_{out}^{node} \boldsymbol{I_j}$$

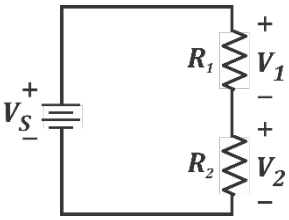
EXAMPLE:

$$I_1 + I_2 = I_3$$

VOLTAGE DIVIDER EQUATION

$$V_1 = V_S \frac{R_1}{(R_1 + R_2)}$$

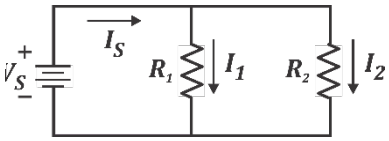
$$V_2 = V_S \frac{R_2}{(R_1 + R_2)}$$



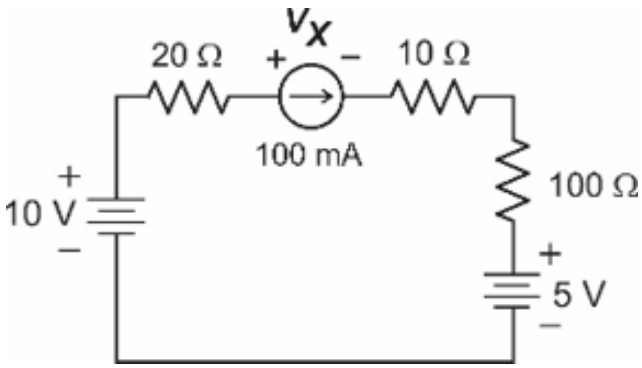
CURRENT DIVIDER EQUATION

$$I_1 = I_S \frac{R_2}{(R_1 + R_2)}$$

$$I_2 = I_S \frac{R_1}{(R_1 + R_2)}$$



P1 KVL

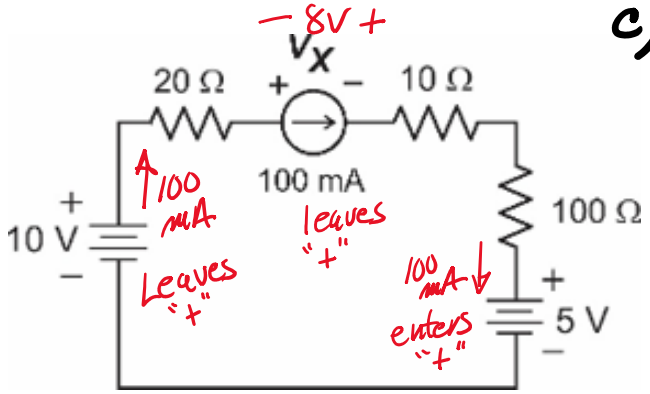


A) DROPS

B) $-10V + 20\Omega(100mA) + V_X + 10\Omega(100mA) + 100\Omega(100mA) + 5V = 0$

$-10V + 2V + V_X + 1V + 10V + 5V = 0$

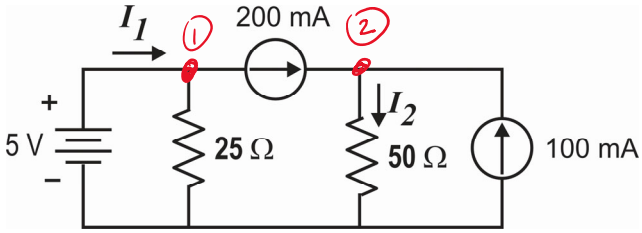
$\boxed{V_X = -8V}$



C)

Component	Power (watts)	Delivered/Absorbed?
10 V	1	Del.
100 mA	0.8	Del.
5 V	0.5	ABS

P2 KCL



A) $\Sigma I_{in} = 0$

B) at NODE (2)

$\Sigma I_{in} = 0$

$200mA + 100mA - I_2 = 0$

$\boxed{I_2 = 300mA}$

The 5V and 25Ω are in parallel

Therefore $I_{25\Omega} = \frac{5V}{25\Omega} = 200mA$

at NODE (1)

$I_1 - I_{25\Omega} - 200mA = 0$

$I_1 - 200mA - 200mA = 0$

$\boxed{I_1 = 400mA}$

C) $P_{5V} = VI = (5V)(400mA)$

$= 2W$ Delivered

I_1 current leaves "+"

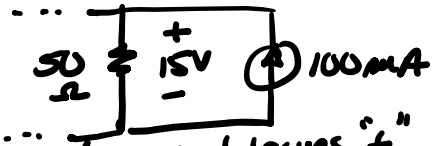
D) $P_{100mA} = VI$

50Ω in parallel with 100mA

$V_{50\Omega} = (I_2)(150\Omega) = 15V$

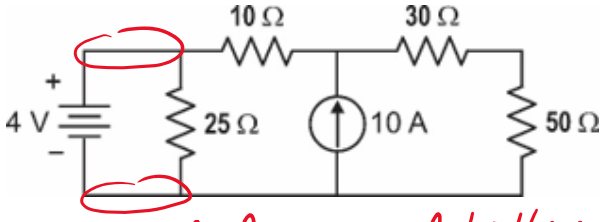
$P_{100mA} = (15V)(100mA)$

$= 1.5W$ Delivered



100mA current leaves "+"

A)

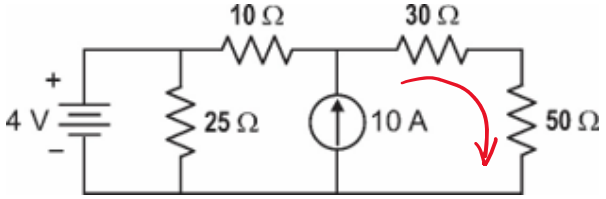


connected top and bottom

Parallel

4V, 25Ω

B)

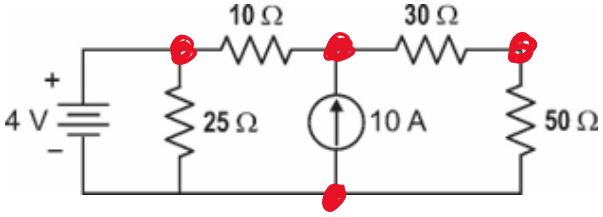


same current

Series

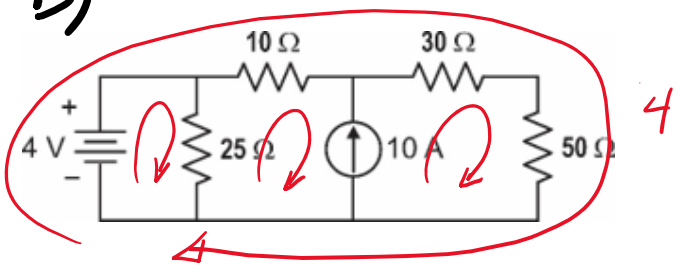
30Ω, 50Ω

C)

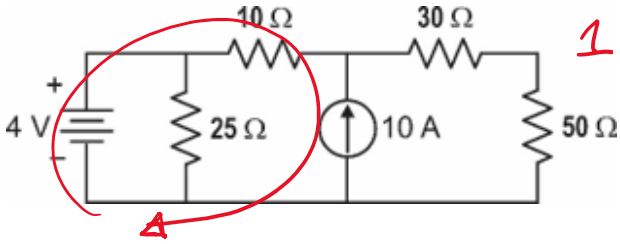


4 NODES

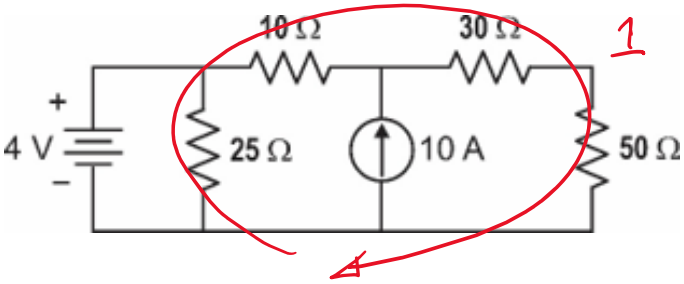
D)



4



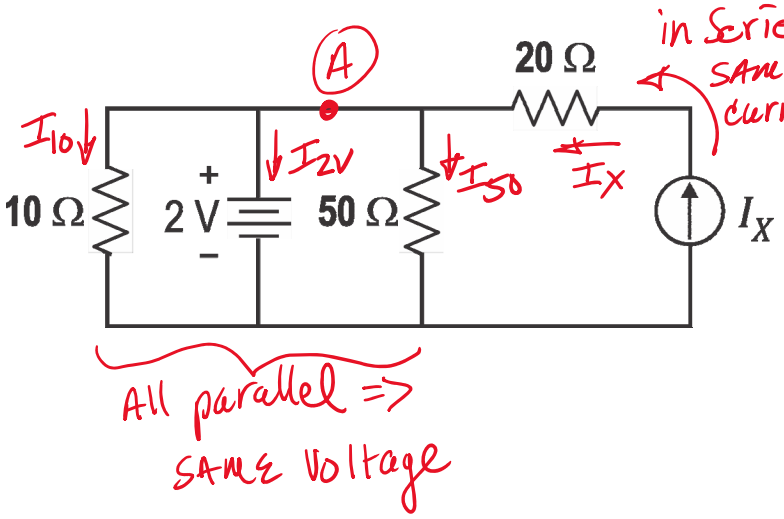
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6 Loops

P4 Extra Credit



$I_{50} = \frac{2V}{50\Omega} = 40mA$

$I_{10} = \frac{2V}{10\Omega} = 200mA$

for no power in 2V

$I_{2V} = 0$

KCL at A:

$\sum_{in} I = 0$

$-200mA - 40mA - I_{2V} + I_X = 0$

$\boxed{I_X = 240mA}$